

DIAL ALERT Data Collection and Understanding

Dataset overview and complete data dictionary

1,072	165,986	4.37M	106,758
PUBLIC PATIENT RECORDS	SESSION ROWS	MONITOR RECORDS	ELIGIBLE SESSIONS

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PROJECT	DIAL ALERT Early Warning for Intradialytic Hypotension
SOURCE DATASET	HEMOBP Version 3
REPORT DATE	September 2026

ACADEMIC DECISION SUPPORT STUDY NOT FOR CLINICAL CARE

This report documents dataset provenance, structure, completeness, validity checks, cohort construction, and variable definitions before model development. It does not establish clinical safety or readiness for patient care.

Executive Summary

DIAL ALERT uses HEMOBP Version 3, a public, de-identified hemodialysis dataset released on Figshare under CC BY 4.0 and described in Scientific Data. The source combines patient characteristics, session-level treatment records, and time-stamped blood-pressure and dialysis-machine measurements. This longitudinal structure is well aligned with the proposed classification task: estimate, at the start of a dialysis session, the probability of a later blood-pressure-defined intradialytic hypotension event. [1,2]

The downloaded public files contain 1,072 patient records, 165,986 session rows, and 4,366,298 monitor records. After deterministic patient-date linkage and preregistered eligibility checks, the analysis-ready table contains 106,758 sessions from 830 patients. The primary label occurs in 9,067 sessions, a prevalence of 8.493%. All counts in this report come from the actual files, not from values copied from the publication.

The dataset is strong enough for a rigorous academic capstone, but it is not a deployment-grade clinical dataset. It is retrospective and single-center; the public files do not contain symptoms, interventions, medications, laboratory results, or broad comorbidity profiles. Accordingly, the primary outcome is explicitly described as BP-defined IDH rather than a complete symptomatic clinical diagnosis.

830	40	8.493%	CC BY
ANALYTIC PATIENTS	ANALYTIC VARIABLES	PRIMARY EVENT RATE	DATA LICENSE

Rubric Evidence

Rubric expectation	Evidence in this deliverable
High-quality dataset and source cited	Peer-reviewed data descriptor, DOI-stable Figshare release, license documented, and file hashes verified.
Comprehensive dataset overview	Grain, dimensions, date coverage, linkage keys, cohort flow, missingness, duplicates, ranges, and outlier policy are reported.
Complete data dictionary	All 23 raw fields and all 40 analysis-table fields are defined with type, unit, values, missingness, meaning, and analytic role.
Reproducibility	Raw files are immutable; transformation rules are code-based; checksums, paths, and commands are documented.

Dataset Selection and Justification

HEMOBP was selected because its three linked tables support the exact temporal structure required by DIAL ALERT. Patient characteristics and predialysis measurements are available before or at the index time, while subsequent blood-pressure observations can define the outcome. This separation permits an auditable prediction-time boundary and reduces the risk of post-index data leakage.

- Industry relevance: hemodialysis is a repeated, safety-critical treatment process in which hypotension risk must be assessed session by session.
- Modelling relevance: the data support supervised binary classification, longitudinal history features, class-imbalance analysis, explainability, and patient-grouped validation.
- Scale: more than four million monitor records provide enough repeated observations for robust preprocessing and model comparison.
- Transparency: the peer-reviewed descriptor and versioned public archive make the acquisition route, license, and data-generating context traceable.

Source and Provenance

File	Grain	Rows x columns	Coverage	Size	Verified MD5
idp.csv	One row per patient	1,072 x 5	First dialysis 1990-01 to 2016-02	0.03 MB	31d269f59bf4acc719f392ad9164d08a
d1.csv	One row per recorded dialysis session	165,986 x 8	2013-06-01 to 2018-05-31	10.62 MB	7db3b48bae2c73e3ecd731dbcd452233
vip.csv	One row per time-stamped dialysis monitor state	4,366,298 x 10	2013-06-01 14:56:40 to 2018-07-01 06:22:57	250.67 MB	ff1589610cf7aa3b3e01ba979b6bdc44

Integrity result. Each local MD5 checksum exactly matches the checksum published in the Figshare Version 3 metadata. This verifies that the audited files are byte-for-byte copies of the cited public release.

Source-count discrepancy. The Scientific Data article reports 1,075 outpatients, whereas the released Idp file contains 1,072 rows and 1,072 unique patient identifiers, a difference of three. No undocumented records were reconstructed. All analyses use the public file contents, and the discrepancy remains an explicit provenance limitation. [1,2]

Collection Context and Governance

Element	Documented context
Care setting	Retrospective outpatient chronic hemodialysis care at a tertiary center in Taiwan
Observation window	June 2013 to July 2018 across the released tables
Ethics record	Institutional review board identifier 16MMHIS044, as reported in the data descriptor
Privacy	De-identified patient identifiers; dates and times retained for longitudinal linkage
Reuse conditions	Figshare Version 3 distributed under Creative Commons Attribution 4.0

HEMOBP source tables and DIAL ALERT analytic unit

Repeated monitor records are reduced to one leakage-controlled row per eligible dialysis session.

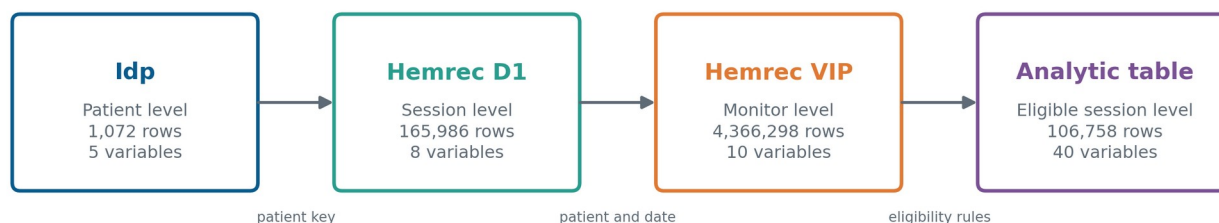


Figure 1. Three source tables are linked by de-identified patient ID and calendar date, then collapsed to one prediction row per eligible dialysis session.

Data Architecture and Linkage

The source has three different grains. **Idp** contains one row per patient. **Hemrec D1** contains one row per recorded dialysis session. **Hemrec VIP** contains repeated, time-stamped monitor states within a session. The analytic unit is one eligible patient-session, not one monitor reading and not one patient.

1. Normalize D1 and VIP dates and link records on de-identified patient ID plus calendar date.
2. Use the earliest valid active-dialysis VIP record within the first 30 elapsed minutes as the index record. Active treatment requires blood flow greater than zero.
3. Require index SBP from 90 to 200 mmHg, index DBP from 30 to 150 mmHg, at least two later distinct elapsed-minute values, and follow-up to at least minute 120.
4. Derive the primary outcome only from records after the index observation. Retain outcome and post-treatment fields for audit, but exclude them from predictors.
5. Derive prior-session features with chronological shifts and cumulative calculations so the current or future session cannot enter its own history.

Linkage assumption. Calendar date is the shared session key available in the public tables. Duplicate patient-date D1 rows are resolved deterministically after stable ordering. If a patient had two distinct treatments on one calendar date, this key could conflate them; that risk cannot be fully tested because no public session ID is provided.

Sampling pattern. VIP observations are not perfectly regular. The paper notes routine blood-pressure collection at approximately 30-minute intervals with additional measurements according to clinical need. Measurement density may therefore contain information about clinician concern and should not be treated as random. [1]

Feature Type Summary

Data type	Raw fields	Analytic fields
Binary	1	5
Categorical	1	1
Date	1	2
Datetime	2	1
Integer	2	5
Numeric	11	23
String	3	1
Time	2	2

Cohort Construction

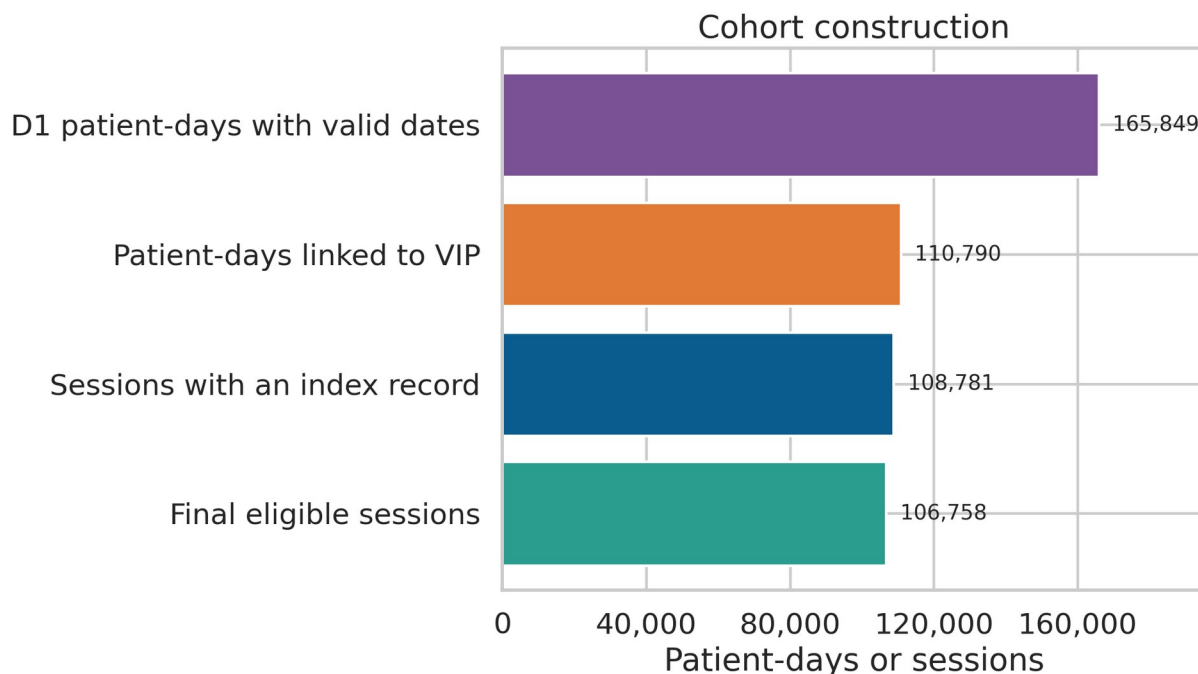


Figure 2. Deterministic cohort flow from the three public source files to 106,758 eligible session-level observations.

Stage	Count	Rule or interpretation
Patient records in ldp	1,072	Patient roster available for linkage
D1 raw session rows	165,986	Includes 128 rows without a usable session date
Valid unique patient-days in D1	165,849	After invalid-date removal and patient-date deduplication
VIP rows linked to D1	2,228,411	Monitor rows with a matching patient-date session
Linked patient-days	110,790	At least one linked VIP record
Patient-days with valid index	108,781	Earliest active record within 30 minutes and valid index BP
Final eligible sessions	106,758	Baseline SBP at least 90 and adequate post-index follow-up

Selection implication. The final cohort includes 830 of the 1,072 public patients. Excluding baseline hypotension, short or sparsely measured sessions, and unlinked patient-days improves label validity but may make the analytic cohort systematically healthier, more observable, or operationally different from excluded sessions. External validation is therefore required before clinical use.

Data Quality and Missingness

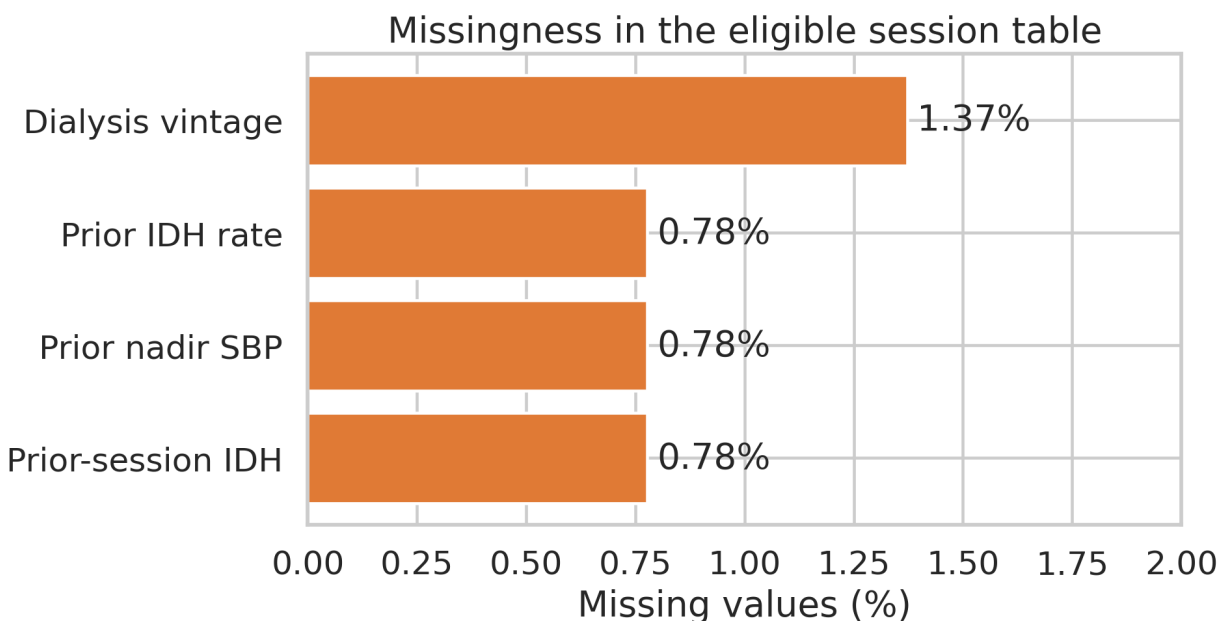


Figure 3. Missingness is concentrated in 128 D1 rows and in history or dialysis-vintage variables in the final analytic table.

Table	Variable group	Missing n	Missing percent	Interpretation
Idp	All 5 variables	0	0.000%	No source missingness detected
Hemrec D1	All fields except pid	128 each	0.077%	The same 128 rows lack all session details
Hemrec VIP	All 10 variables	0	0.000%	No encoded nulls detected
Analytic	dialysis_vintage_years	1,465	1.372%	Unavailable or invalid chronology
Analytic	prior_session_idh	830	0.777%	First eligible session per patient
Analytic	prior_nadir_sbp	830	0.777%	First eligible session per patient
Analytic	prior_idh_rate	830	0.777%	First eligible session per patient
Analytic	Remaining 36 variables	0	0.000%	Complete after eligibility filters

- Null handling is rule-based. D1 rows without a session date cannot be linked and are excluded before cohort construction.
- History-feature missingness is structural, not evidence of poor source quality: the first eligible session has no earlier eligible session.
- Dialysis-vintage missingness remains explicit and will be handled inside the modelling pipeline, fitted on training data only.
- No complete-case deletion is applied merely because a predictor is missing; the modelling stage will compare transparent imputation and missing-indicator strategies.

Duplicates and Referential Integrity

Table	Finding	Disposition
Idp	0 exact duplicates; 0 duplicate patient IDs	Patient key is unique
Hemrec D1	5 exact duplicates; 9 duplicate patient-date rows	Stable order retained and first row kept for one session per patient-date

Table	Finding	Disposition
Hemrec VIP	0 exact duplicates after linkage and validity screening	No linked analytic monitor row removed as an exact duplicate
Analytic table	One row per eligible patient-date	Patient-date is unique by construction

Outliers and Plausibility

Outlier detection separates validity from rarity. Clearly unusable values are screened through predefined physiological and operational bounds before index selection. The interquartile-range rule is then used only to describe unusual but potentially valid observations; it is not used as an automatic deletion rule. This protects clinically important extremes and prevents data-dependent trimming from silently changing the target population.

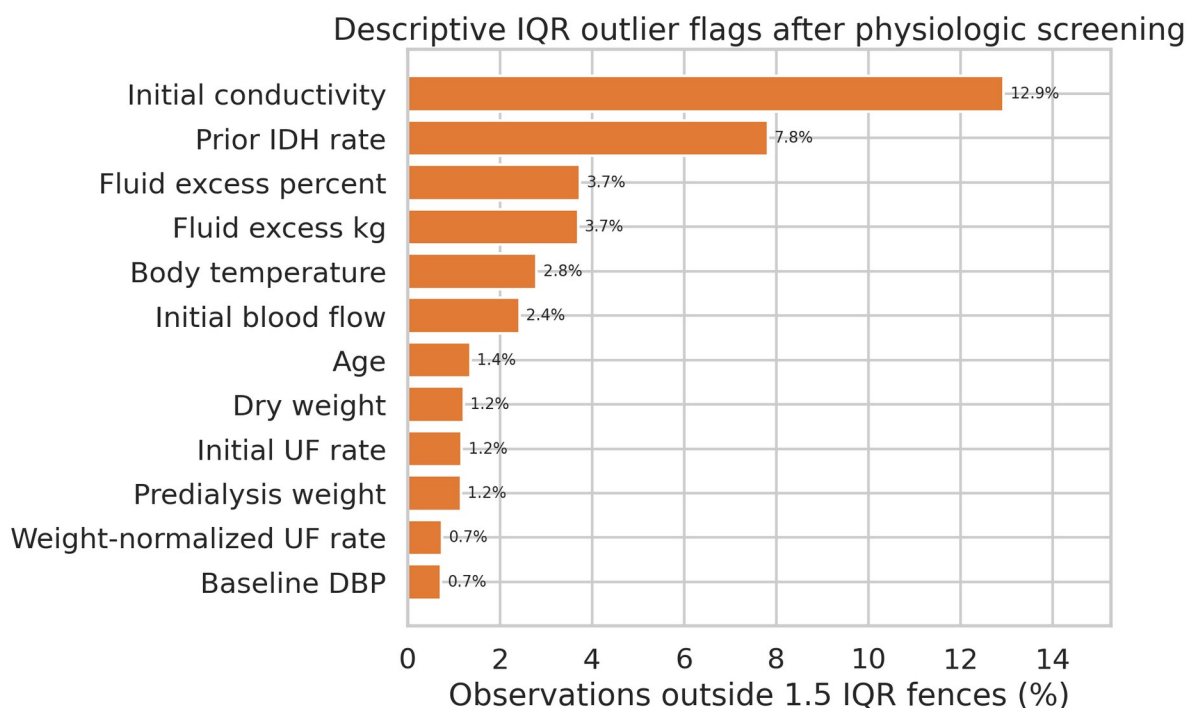


Figure 4. IQR flags among candidate predictors. High flag rates can reflect narrow central distributions rather than invalid records.

Feature	Min	Median	Max	IQR flags	Flag rate
initial_conductivity_ms_cm	10.00	14.00	20.00	13,813	12.94%
prior_idh_rate	0.00	0.04	1.00	8,275	7.81%
fluid_excess_pct	-28.74	3.93	64.95	3,987	3.73%
fluid_excess_kg	-23.80	2.20	37.70	3,945	3.70%
temperature	35.00	36.40	39.50	2,981	2.79%
initial_blood_flow_ml_min	2.00	210.00	400.00	2,589	2.43%
age_years	18.00	66.00	94.00	1,450	1.36%
dryweight	30.00	55.30	136.00	1,294	1.21%
initial_uf_l_h	0.00	0.60	3.00	1,247	1.17%
weightstart	30.60	57.80	172.70	1,232	1.15%

Priority quality checks. Conductivity has the largest IQR flag rate, 12.94%, because most readings are tightly concentrated near 14 mS/cm. Derived fluid-excess values range from -23.8 to 37.7 kg, and normalized ultrafiltration reaches 92.88 mL/kg/hour. These values are retained for transparency but will receive sensitivity checks, robust scaling or winsorization fitted only on training data, and individual-record review where feasible.

Outcome Definition and Dataset Fitness

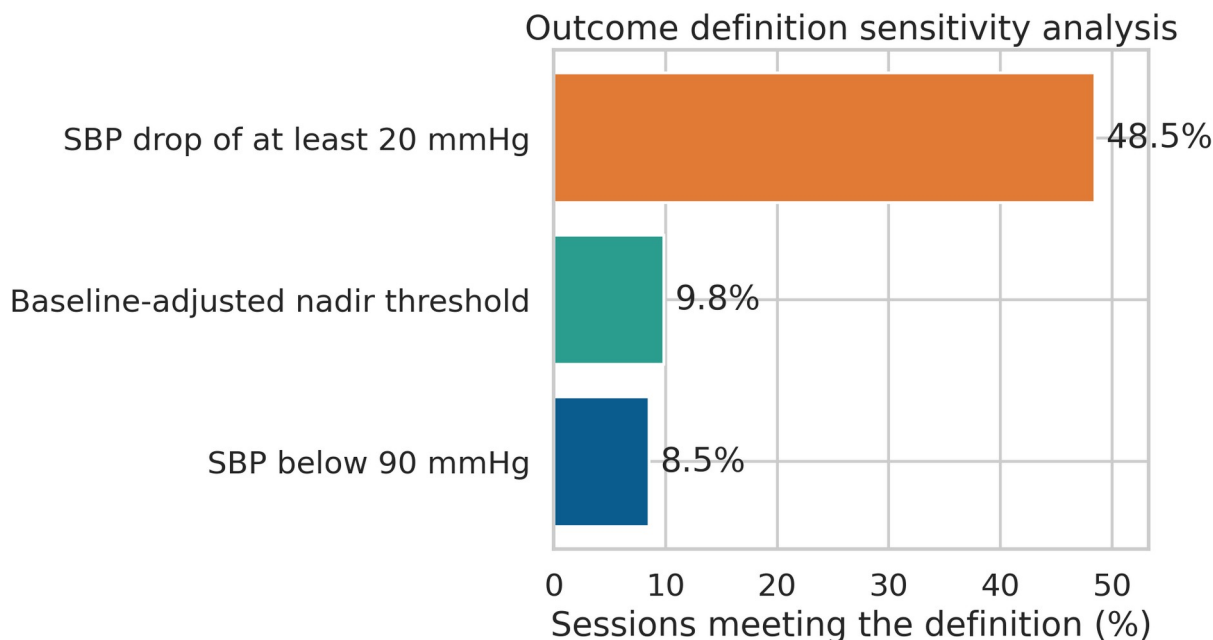


Figure 5. Outcome prevalence varies substantially by blood-pressure definition, reinforcing the need to preregister a primary label and report sensitivity analyses.

Status	Variable	Operational definition	Events	Prevalence
Primary	idh_absolute	Any later SBP below 90 mmHg	9,067	8.493%
Sensitivity	idh_flythe	Later nadir below 90 if baseline below 160; otherwise below 100	10,508	9.843%
Sensitivity	idh_drop_20	Any later SBP at least 20 mmHg below baseline	51,753	48.477%

Fitness conclusion. HEMOBP is fit for an academic study of early, BP-defined risk prediction. It is not sufficient to claim prediction of the full clinical syndrome of IDH because symptoms and interventions are absent. It also does not support causal claims about treatment settings: dialysate temperature, conductivity, ultrafiltration, and blood flow were assigned in routine care rather than randomized.

Known Limitations and Controls

Limitation	Why it matters	Planned control
Single center and country	Transportability to other dialysis units, devices, and populations is unknown.	Use patient-grouped internal validation and require independent external validation.
Repeated sessions	A random row split would place the same patient in training and testing.	Use patient-grouped splits and report patient-level bootstrap uncertainty.
Outcome incompleteness	No symptoms or rescue interventions are available.	Name the label BP-defined IDH and present alternate BP definitions.

Limitation	Why it matters	Planned control
Informative sampling	Extra BP readings may occur when clinicians are concerned.	Exclude post-index count features from prediction and discuss measurement-process bias.
Limited clinical covariates	Most laboratory, medication, and comorbidity data are absent.	Constrain claims and document omitted-variable risk.
Sensitive attributes	Public data include recorded sex and age but not race or socioeconomic status.	Audit available subgroups and state that unobserved-group fairness cannot be assessed.

Reproducibility Record

The source CSV files remain immutable in `data/raw`. The session-level table is generated by code, and every exclusion and feature definition is encoded rather than performed manually. The workbook supplied with this report contains the full audit fields and can be filtered by source table, modelling role, or prediction-time availability.

Purpose	Command or path
Build analytic table	<code>python src/build_session_dataset.py</code>
Regenerate audit and figures	<code>python src/generate_step2_assets.py</code>
Regenerate dictionary workbook	<code>node src/create_step2_dictionary_workbook.mjs</code>
Regenerate this report	<code>python src/create_step2_report.py</code>
Analysis-ready dataset	<code>data/processed/hemobp_session_level.csv.gz</code>
Machine-readable audit	<code>artifacts/step2_dataset_audit.json</code>

References

- [1] Lin CJ, Chen YY, Pan CF, Wu V, Wu CJ. Dataset supporting blood pressure prediction for the management of chronic hemodialysis. *Scientific Data*. 2019;6:313. <https://doi.org/10.1038/s41597-019-0319-8>
- [2] Chien CY. HEMOBP. Figshare. Version 3. Dataset. 2019. CC BY 4.0. <https://doi.org/10.6084/m9.figshare.6260654.v3>

Data Dictionary Conventions

The appendices list every field in the three public source files and every field in the session-level analytic table. Observed ranges describe this specific release; they are not universal physiological limits. Missing percentages use the row count of the relevant table as the denominator. Role and availability state whether a field may be used by a model at the index time. The companion spreadsheet preserves additional provenance and quality-note columns for all 63 variables.

Dictionary element	Interpretation
Variable	Exact column name used in the corresponding CSV file
Type	Logical data type after parsing; source files may store values as text
Unit	Physical or coding unit; source-defined index is used where the publication does not document a unit
Values or observed range	Allowed codes for categories or minimum-to-maximum values observed in the audited release
Missing	Number and percentage of rows encoded as null after parsing
Role and availability	Purpose in the analytic design and whether it exists before, at, or after prediction time

Appendix A Raw Source Data Dictionary

The raw dictionary contains 23 variables across Idp, Hemrec D1, and Hemrec VIP. Raw values are never overwritten.

Idp

Variable	Type	Unit	Values or observed range	Missing	Definition	Analytic role
pid	string	none	1,072 unique de-identified IDs	0 (0.000%)	De-identified patient identifier	Primary patient key
gender	categorical	none	F or M	0 (0.000%)	Recorded sex field	Patient characteristic and fairness attribute
birthday	integer	year	1920 to 1998	0 (0.000%)	Recorded birth year	Source for age derivation
first_dialysis	date	year-month	1990-01 to 2016-02	0 (0.000%)	Month of first hemodialysis treatment	Source for dialysis vintage
DM	binary	0 or 1	0 = no diabetes; 1 = diabetes	0 (0.000%)	Recorded diabetes status	Patient characteristic

Hemrec D1

Variable	Type	Unit	Values or observed range	Missing	Definition	Analytic role
pid	string	none	1,072 unique de-identified IDs	0 (0.000%)	De-identified patient identifier	Patient linkage key
keyindate	datetime	timestamp	2013-06-01 to 2018-05-31	128 (0.077%)	Date and time used to identify the dialysis session	Session linkage key
dialysisstart	time	HH:MM	24-hour clock time as stored	128 (0.077%)	Recorded dialysis start clock time	Session audit field
dialysisend	time	HH:MM	24-hour clock time as stored	128 (0.077%)	Recorded dialysis end clock time	Post-session audit field
weightstart	numeric	kg	30.6 to 172.7	128 (0.077%)	Body weight before dialysis	Baseline predictor source
weightend	numeric	kg	30 to 164.1	128 (0.077%)	Body weight after dialysis	Post-session field
dryweight	numeric	kg	30 to 136	128 (0.077%)	Prescribed dry weight	Baseline predictor source
temperature	numeric	degrees Celsius	35 to 39.6	128 (0.077%)	Body temperature recorded for the session	Baseline predictor source

Hemrec VIP

Variable	Type	Unit	Values or observed range	Missing	Definition	Analytic role
pid	string	none	1,072 unique de-identified IDs	0 (0.000%)	De-identified patient identifier	Patient linkage key
datetime	datetime	timestamp	2013-06-01 14:56:40 to 2018-07-01 06:22:57	0 (0.000%)	Timestamp of the monitor record	Patient-date linkage and ordering
measuretime	integer	source-defined index	0 to 98	0 (0.000%)	Blood-pressure measurement index as stored	Ordering audit; source unit not documented
sbp	numeric	mmHg	31 to 200	0 (0.000%)	Systolic blood pressure	Index predictor and later outcome source
dbp	numeric	mmHg	30 to 192	0 (0.000%)	Diastolic blood pressure	Index predictor source
dia_temp_value	numeric	degrees Celsius	33 to 40	0 (0.000%)	Dialysate temperature setting	Index predictor source
conductivity	numeric	mS/cm	10 to 20	0 (0.000%)	Dialysate conductivity, a proxy for dialysate sodium concentration	Index predictor source
uf	numeric	L/hour	0 to 3	0 (0.000%)	Ultrafiltration rate	Index predictor source
blood_flow	numeric	mL/min	0 to 400	0 (0.000%)	Blood-flow rate through the extracorporeal circuit	Index predictor source
time	numeric	minutes	0 to 370	0 (0.000%)	Elapsed dialysis time	Index selection and later outcome window

Interpretation note. In the source documentation, measuretime is an undocumented measurement index, whereas time is elapsed dialysis time in minutes. They are not treated as interchangeable. Raw minimum and maximum values describe this release; separate index-eligibility checks define the analysis-ready cohort.

Appendix B Analytic Data Dictionary

The analytic table contains 40 variables. Fields marked audit, eligibility, outcome, or post-index are retained for traceability but are excluded from model predictors.

Identifiers and Index Documentation

Variable	Type	Unit	Values or observed range	Missing	Definition	Role and availability
pid	string	Identifier	830 unique IDs in the analytic cohort	0 (0.000%)	De-identified patient identifier	Grouping and split only; never a predictor; Identifier only
session_date	date	Date	2013-06-01 to 2018-05-31	0 (0.000%)	Date used to link a dialysis session across source tables	Identifier and temporal audit; Audit only
index_datetime	datetime	Timestamp	2013-06-01 19:53:40 to 2018-05-31 20:41:17	0 (0.000%)	Timestamp of the earliest valid active-dialysis record within 30 minutes	Index-time documentation; Audit only
index_minute	numeric	minutes	0 to 30	0 (0.000%)	Elapsed dialysis minute at the index record	Cohort audit; not a predictor; Audit only

Index and Predialysis Predictors

Variable	Type	Unit	Values or observed range	Missing	Definition	Role and availability
baseline_sbp	numeric	mmHg	90 to 200	0 (0.000%)	Systolic BP at the index record	Predictor; Known at index
baseline_dbp	numeric	mmHg	30 to 150	0 (0.000%)	Diastolic BP at the index record	Predictor; Known at index
initial_dialysate_temp_c	numeric	degrees Celsius	35 to 39	0 (0.000%)	Dialysate temperature at the index record	Predictor; Known at index
initial_conductivity_ms_cm	numeric	mS/cm	10 to 20	0 (0.000%)	Dialysate conductivity at the index record	Predictor; Known at index
initial_uf_l_h	numeric	L/hour	0 to 3	0 (0.000%)	Ultrafiltration rate at the index record	Predictor; Known at index
initial_blood_flow_ml_min	numeric	mL/min	2 to 400	0 (0.000%)	Blood-flow rate at the index record	Predictor; Known at index
weightstart	numeric	kg	30.6 to 172.7	0 (0.000%)	Body weight before dialysis	Predictor; Known before treatment
dryweight	numeric	kg	30 to 136	0 (0.000%)	Prescribed dry weight	Predictor; Known before treatment
temperature	numeric	degrees Celsius	35 to 39.5	0 (0.000%)	Body temperature recorded for the session	Predictor; Known at session start
gender	categorical	F or M	F or M	0 (0.000%)	Recorded sex field in the source data	Predictor and fairness audit; Known before treatment
DM	binary	0 or 1	0 = no diabetes; 1 = diabetes	0 (0.000%)	Recorded diabetes status	Predictor and subgroup audit; Known before treatment

Variable	Type	Unit	Values or observed range	Missing	Definition	Role and availability
age_years	numeric	years	18 to 94	0 (0.000%)	Session year minus birth year	Predictor and fairness audit; Known before treatment
dialysis_vintage_years	numeric	years	0 to 27.28	1,465 (1.372%)	Time from first dialysis to session date	Predictor; Known before treatment
fluid_excess_kg	numeric	kg	-23.8 to 37.7	0 (0.000%)	Predialysis weight minus dry weight	Engineered predictor; Known before treatment
fluid_excess_pct	numeric	percent of dry weight	-28.74 to 64.95	0 (0.000%)	Fluid excess divided by dry weight	Engineered predictor; Known before treatment
initial_uf_ml_kg_h	numeric	mL/kg/hour	0 to 92.88	0 (0.000%)	Initial UF rate normalized by predialysis weight	Engineered predictor; Known at index
baseline_map	numeric	mmHg	50.67 to 165.67	0 (0.000%)	DBP plus one-third of pulse pressure	Engineered predictor; Known at index
baseline_pulse_pressure	numeric	mmHg	5 to 162	0 (0.000%)	Baseline SBP minus DBP	Engineered predictor; Known at index

Prior Session Predictors

Variable	Type	Unit	Values or observed range	Missing	Definition	Role and availability
prior_session_idh	binary	0 or 1	0 = no; 1 = yes	830 (0.777%)	Primary outcome in the immediately preceding observed session	History predictor; earlier sessions only; Prior sessions only
prior_nadir_sbp	numeric	mmHg	37 to 198	830 (0.777%)	Later nadir SBP in the preceding observed session	History predictor; earlier sessions only; Prior sessions only
prior_session_count	integer	sessions	0 to 735	0 (0.000%)	Number of earlier eligible sessions for the patient	History predictor; log transformed; Prior sessions only
prior_idh_rate	numeric	0 to 1	0 to 1	830 (0.777%)	Cumulative BP-defined IDH rate across earlier sessions	History predictor; earlier sessions only; Prior sessions only

Outcome and Post Index Audit

Variable	Type	Unit	Values or observed range	Missing	Definition	Role and availability
later_records	integer	records	2 to 1,364	0 (0.000%)	Number of records after the index measurement	Outcome-quality audit; not a predictor; Post-index audit only
later_distinct_minutes	integer	distinct minutes	2 to 242	0 (0.000%)	Distinct elapsed-time values after the index measurement	Eligibility; not a predictor; Post-index eligibility only
last_observed_minute	numeric	minutes	120 to 370	0 (0.000%)	Latest elapsed dialysis time recorded after index	Eligibility; not a predictor; Post-index eligibility only

Variable	Type	Unit	Values or observed range	Missing	Definition	Role and availability
nadir_sbp	numeric	mmHg	37 to 198	0 (0.000%)	Lowest SBP recorded after index	Outcome derivation; never a predictor; Post-index outcome derivation only
mean_later_sbp	numeric	mmHg	64.74 to 198.67	0 (0.000%)	Mean SBP after index	Outcome description; never a predictor; Post-index outcome description only
idh_absolute	binary	0 or 1	0 = no; 1 = yes	0 (0.000%)	Later SBP below 90 mmHg	Primary outcome; Primary outcome
idh_drop_20	binary	0 or 1	0 = no; 1 = yes	0 (0.000%)	Later SBP at least 20 mmHg below baseline	Sensitivity outcome; Sensitivity outcome
idh_flythe	binary	0 or 1	0 = no; 1 = yes	0 (0.000%)	Later nadir below 90 mmHg when baseline is below 160, otherwise below 100	Sensitivity outcome; Sensitivity outcome
dialysisstart	time	Clock time	24-hour clock time as stored	0 (0.000%)	Recorded dialysis start time	Audit; not a predictor; Audit only
dialysisend	time	Clock time	24-hour clock time as stored	0 (0.000%)	Recorded dialysis end time	Post-index field; excluded; Post-index field; excluded
weightend	numeric	kg	30 to 164.1	0 (0.000%)	Body weight after dialysis	Post-index field; excluded; Post-treatment field; excluded
birthday	integer	year	1,920 to 1,998	0 (0.000%)	Recorded birth year	Used to derive age; excluded directly; Source only; replaced by age
first_dialysis	date	Year and month	1990-01-01 to 2016-01-01	0 (0.000%)	Recorded month of first dialysis	Used to derive vintage; Source only; replaced by vintage
session_year	integer	year	2,013 to 2,018	0 (0.000%)	Calendar year of the session	Temporal audit; excluded from model; Temporal audit only